

Two Slit Interference, One Photon at a Time Experiment Outline

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ABSTRACT

It is well known that quantum behavior contradicts classical physics in profound ways, one example being the particle-wave duality of light and other particles. We explore this phenomena by reproducing the famous two-slit experiment, then repeating the experiment with a light source so dim that only one photon travels through the apparatus at a time. We measure the interference pattern of both experiments finding the same qualitative behavior. We fit Fraunhofer intensity equations to our data to calculate the slit widths a and separation b , and find: $a = 0.071 \pm 0.001$ mm and $b = 0.410 \pm 0.005$ mm. We use these measurements to show that the single-photon interference pattern produces the same shape with a reduced χ^2 fit of 39.4, indicating that the particle has a spatial extent allowing it to be disrupted by the second slit and interfere with itself. This experiment relies on the particle-like nature of light but this result is strong evidence for its wave-like nature, thus demonstrating the concept of duality.

1 INTRODUCTION

The nature of light was debated for centuries before arriving at the modern understanding of its paradoxical behavior. In the nineteenth century, Young’s famous double-slit experiment [Young \(1804\)](#) seemingly proved that light behaved as a wave because it creates interference patterns when passed through a narrow opening, or slit. This conjecture was further supported when Maxwell showed that light is produced from oscillating electric and magnetic waves ([Maxwell 1865](#)). Half a century later, Einstein showed that light exists in discrete packets of energy –photons– supporting the theory for its existence as a particle. We now understand that light simultaneously exhibits both of these behaviors, often described as particle-wave duality.

Analyzing the interference pattern of light sources passed through two slits was imperative in understanding its properties. Even more interesting, the same result can be shown when a single photon passes through the double slit at a time. In this case, if the photon behaves as a particle it must pass through one slit or the other, and we expect to see one peak behind each of the two slits. Behaving as a wave, a single photon could simultaneously pass through both slits, interfering with itself and creating the same interference pattern as when multiple particles are streaming through the slits. Here, we test the extent to which these hypotheses are correct and discuss the implications for our understanding of quantum mechanics.

In Section 2 we present the theory behind the interference patterns for both the particle and wave hypothesis. In Section ?? we detail our experimental set up and procedure for recording the photon detections. In Section 4 we present our data and show the interference patterns. We discuss these findings in Section 5.

2 THEORY

We aim to analyze the pattern created when photons are passed through a single or double slit arrangement. We first shine a laser through the apparatus shown in Figure 1. The photons travel to the right in the box, reaching a double slit whose separation is of order

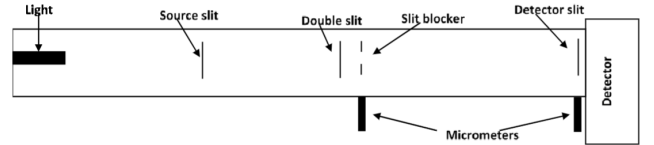


Figure 1. Diagram of experiment apparatus. At first the light source is a laser and the detector is a photo-diode. We then switch the light source to a dim light bulb and the detector to a photomultiplier tube. The micrometers control the position of the slit blocker and detector slit.

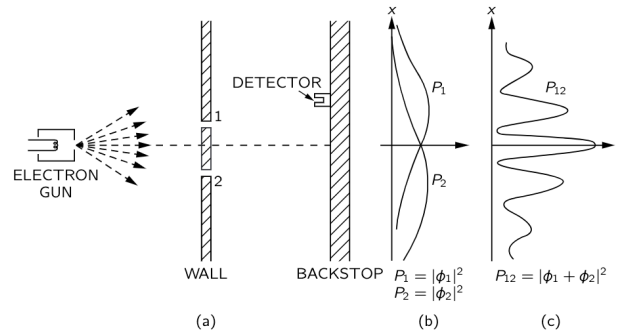


Figure 2. Experiment schematic with probability distributions of particle detections. The curves in panel b show the probability functions expected when the photons pass through only one of the slits, and the curve in panel c shows the interference pattern when they pass through both slits.

the wavelength of the photons. If photons behave as a particle, then each photon passes through either slit 1 or slit 2. In this case, a measurement of the intensity over a long period of time is expected to resemble those in panel b of Figure 2 ([Feynman 2006](#)). However, the actual intensity pattern is shown in panel c, which is the prediction

for wavelike behavior. The central peak occurs where the photon wavelengths constructively interfere, and the troughs result from destructive interference. If the probability amplitude of each slit is given by the imaginary number Φ , then we see that the intensity is not given by $P = P_1 + P_2 = |\Phi_1|^2 + |\Phi_2|^2$ as expected if the photon behaved as a particle, but rather is given by $P = |\Phi_1 + \Phi_2|^2$. We aim to measure this function in this experiment.

We start with the Fraunhofer intensity equations for single and double slit interference patterns, respectively:

$$I(\theta) = I(0) \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad (1)$$

$$I(\theta) = 4I(0) \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2(\beta) \quad (2)$$

where I is the intensity and θ is the angle of the observed position with respect to the center the double slit (Carmen/Canvas 2024). We define $\alpha = (ka/2)\sin(\theta)$ and $\beta = (kb/2)\sin(\theta)$ where a is the width of each slit, b is the spacing between the slits, and $k = 2\pi/\lambda$ where λ is the wavelength of the light source. These equations can be rewritten as

$$I(x) = I_0 \left(\frac{\sin(A(x-x_0))}{A(x-x_0)} \right)^2 \quad (3)$$

and

$$I(x) = 4I_0 \left(\frac{\sin(A(x-x_0))}{A(x-x_0)} \right)^2 \cos^2(B(x-x_0)) \quad (4)$$

where $A = \frac{\pi a}{\lambda D}$ and $B = \frac{\pi b}{\lambda D}$, and D is the distance between the double slit and the screen. The full derivation is given in the appendix.

Fitting the data for A and B then provides an easy calculation of the physical width of each slit and their separation. Holding these variable constant, we can vary the light source and show the interference pattern remains constant.

To verify that only one photon is passing through the double slit at once, we compare the detection rate of the photons to the time each one spends traveling from the light source to the detector. The measured photon arrival rate, R must be corrected by the detector efficiency, η , to account for the photons which arrive undetected. The time for the photon to reach the detector from the light source is L/c where L is the length of the experiment box and c is the speed of light. The probability then that at a random time there is a photon in the box is given by:

$$P = \frac{t_{\text{travel}}}{t_{\text{arrival}}} = \frac{\eta L}{Rc} \quad (5)$$

Because this is the probability that a single photon is passing through the double slit at a time, the probability that two are traveling through simultaneously is P^2 .

3 METHODS

In this section we detail the procedure for measuring photon intensity. In Section 3.1 we describe the experiment using the laser light source and photodiode detector. We then describe the procedure when using the dim light bulb source and photomultiplier tube detector in Section 3.2.

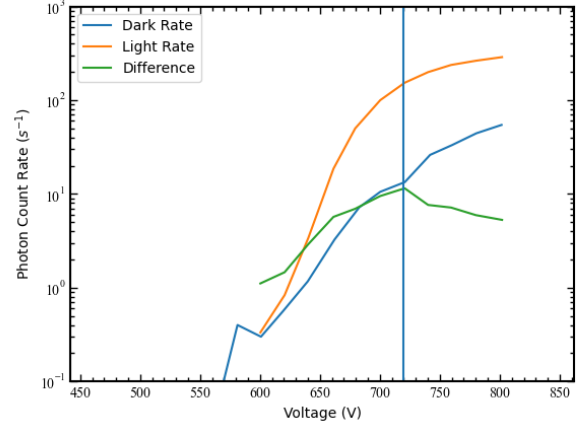


Figure 3. The photon count per second as a function of PMT operation voltage. The dark rate (blue curve) is effectively the background rate, and the light rate (orange curve) is the detection rate of actual photons arriving at the detector. We find the voltage where the difference between these curves (shown by the green curve) is maximized, which occurs at 720.7 V.

3.1 Laser Light Source

We first use the set up shown in Figure 1 where the light source is a laser with wavelength $\lambda = 670 \pm 5$ nm. The slit blocker can be moved to cover the left slit, right slit, or neither slit. We begin with a 1 cm² photodiode detector in the detector box on the far right side. We record a background signal of the photodiode detector to be $S_B = 103$ mV.

Following the procedure detailed in Carmen/Canvas (2003) and Carmen/Canvas (2022) we first block the slit on the left side of the box. We use the micrometers to move the detector a distance of 4 mm, recording the photodiode's voltage readout every 0.10 mm. We then block the right slit and repeat these measurements. The uncertainty on the multimeter reading of the photodiode read out is $0.04\%(\text{value}) \pm 0.09\%(\text{range})$.

We move the slit blocker so that the photons are now free to pass through either slit. To better capture the detail of the interference pattern, we increase the resolution of our measurements to 0.05 mm spacing. We repeat the measurements over the same 4 mm range.

3.2 Dim Light Bulb Light Source

We now use a dim green-filtered light bulb with permissible wavelengths $\lambda = 541 - 551$ nm for the light source. The intensity of this source is so low that it only releases a single photon at once. We therefore require a more sensitive detector than the photodiode from Section 3.1. We use a photomultiplier tube (PMT) for the detector, which is able to detect single photon events. The detector is connected with BNC cables to an oscilloscope to record the amplitude of the analog voltage signal produced by the PMT. The oscilloscope is connected to a TTL counter with a fixed voltage threshold to convert the analog pulse to a digital square wave pulse. The TTL counts each of these square waves, producing a photon count per second.

Before measuring the interference pattern, we test the optimal voltage for the PMT to maximize the signal to noise ratio. We begin by recording the background signal as a function of the voltage at which the PMT is operating. We measure the photon arrival count during a 10 second interval over a range of 560–800 V, recording the

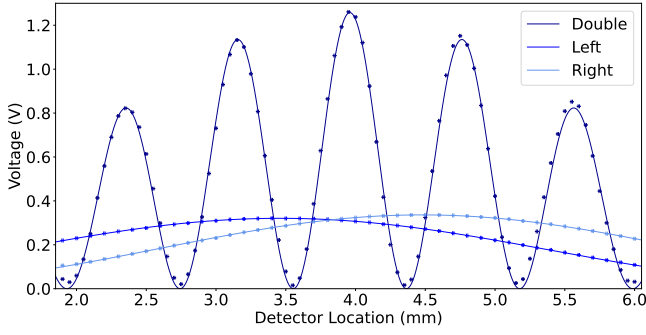


Figure 4. Photodiode voltage reading as a function of detector slit location for when the photons are able to pass through the right slit (light blue), left slit (medium blue), or both slits (dark blue). The error bars are shown with oversized caps. We calculate weighted fits to each of these datasets, which define their respective solid lines. There is an offset between the peaks on the left and right side of the central peak, which occurs because the instrument was imperfectly calibrated.

photon count every 20 V. This defines the "dark rate" shown by the blue curve in Figure 3. We then move the slit blocker so that photons are free to pass through both slits. We repeat the measurements every 20 V over the same range, which defines the "light rate" or orange curve in Figure 3. The green line shows the difference between these counts. To maximize the SNR ratio, we select the voltage where the green curve is maximized, which occurs at $V_{\text{opt}} = 720.7$ V. At this point, the dark rate $R_D = 15.8 \text{ s}^{-1}$. For the remainder of the experiment, the PMT operates at this voltage.

We record photon counts with the slit blocker located in the same three positions described in Section 3.1. We now record the photon count over a 10 second interval at the same 0.10 mm resolution for each of the single slits and 0.05 mm resolution for the double slit arrangement. We measure the photon count over a slightly larger range of detector location, from 0.0 – 7.0 mm.

4 ANALYSIS AND INTERPRETATION

In Section 4.1, we fit the Fraunhofer intensity equations to our data recorded with the laser light source to determine physical characteristics of our experimental set up. We then use these measurements to show that the same interference pattern is produced even when one photon is present in the apparatus at a time in Section 4.2.

4.1 Measuring Physical Slit Sizes

Figure 4 shows our measurements of signal voltage versus detector position when the photons are free to pass through the left slit, right slit, or both slits. The background signal is subtracted from the voltage measurements. The measurement uncertainties come from two primary factors. First is the precision of the multimeter to measure the voltage created by the photodiode, given in Section 3.1. Second, the micrometer is marked every 0.1 mm, thus our precision in reading the slit positions is limited to within 0.05 mm. The error bars for all three data sets are shown with over-sized cap sizes for visibility.

We calculate a weighted least squares fit to each of the single slit data sets. The best fit parameters are given in first section of Table 1. Because multivariate fits tend to give worse estimates for each parameter when the number of variables increases, we report the

Parameter	Value
I_0	$0.34 \pm 0.01 \text{ V}$
left x_0	$3.461 \pm 0.002 \text{ mm}$
right x_0	$4.478 \pm 0.002 \text{ mm}$
A	0.67 ± 0.02
B	3.867 ± 0.007

Table 1. Fit parameters from the data recorded using the laser light source. All values except B are reported from the fit of Equation 4.

best fit parameters with their uncertainties for I_0 , x_0 , and A from Equation 3. Since we have independent fits for when the left or right slit is covered, the values reported are the weighted averages of the two estimates. The uncertainties are calculated from the diagonalized covariance matrix of the best fit parameters for each fit then added in quadrature. The medium blue curve of Figure 4 has a reduced $\chi^2 = 0.002$, and for the light blue curve it is $\chi^2 = 0.003$. This indicates that our error bars are overestimated in both cases, and Equation 3 is an excellent model for the single slit data.

We then use these parameters to initialize a weighted least squares fit of Equation 4 to the double slit data. We report the best fit value for B . The best fit estimate of A only changed by 1% in this new fit. The dark blue curve tracing the interference pattern in Figure 4 has a reduced χ^2 fit of 1.56, suggesting our measurement errors are slightly underestimated. There appears to be a bias in the right peaks where the function underestimates these values. This likely occurs because our voltage readings between the left and right side of the center had a difference of 0.01 V during calibration. Additionally, the behavior in the troughs is poor in every location, indicating a systematic bias in the data. Likely, the detector performs worse at these small voltages, or the multimeter manufacturer has misreported the uncertainty in this range.

We calculate the slit width $a = \lambda DA/\pi$ and their separation $b = \lambda DB/\pi$ using the estimates in Table 1. Our uncertainties are calculated by:

$$\sigma_p^2 = \left(\frac{\partial p}{\partial \lambda} \right) \sigma_\lambda^2 + \left(\frac{\partial p}{\partial D} \right) \sigma_D^2 + \left(\frac{\partial p}{\partial A} \right) \sigma_A^2 \quad (6)$$

where p is the parameter a or b and P is either A or B , respectively. We measure $a = 0.071 \pm 0.001 \text{ mm}$ which is 11σ from the lab manual value of 0.085 (Carmen/Canvas 2003). We also find $b = 0.410 \pm 0.005 \text{ mm}$ which is within 1σ of the expected value (Carmen/Canvas 2003).

4.2 Interference One-Photon-at-a-Time

Figure 5 shows the measurements of photon count over a 10 s interval as a function of the detector position with the dark rate subtracted. Qualitatively, the curves exhibit the same shape as in Figure 4, and the double slit data clearly follows an interference pattern.

We use the estimates of A and B from Section 4.1 to fit weighted least square functions to these new data. The light blue and medium blue curves have reduced χ^2 fits of 0.54 and 0.45, respectively. The dark blue curve has a reduced $\chi^2 = 39.4$. This fit is accurate near the central peak and becomes worse farther out in the wings. The function still misses every trough, suggesting that the dark rate is underestimated, or there is another unaccounted for systematic bias.

However, the general agreement in the qualitative and quantitative characteristics of the pattern suggests that the same interference pattern is produced regardless of the number of photons passing through the slits. From Equation 5, we can calculate the probability that only

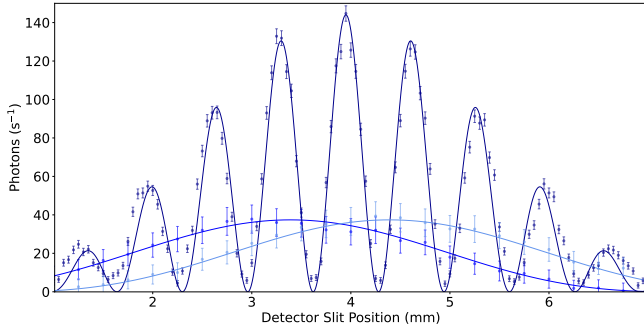


Figure 5. Photon count as a function of detector slit position for the same three cases as Figure 4. The single slit functions still provide excellent fits, but the double slit only agrees with the data near the central peak. The calibration bias remains visible, and the function misses the troughs for every case.

one photon is present in the apparatus at any given time. The detector efficiency η is 0.04 for this PMT [Carmen/Canvas \(2003\)](#). We measure the length from the laser to the detector box to be $L = 1.015 \pm 0.005$ m and the maximum photon arrival rate was measured to be $R = 36$ s^{-1} . This yields a probability of 3.04×10^{-6} that one photon is present at any given time. Therefore the probability that this pattern is *not* produced by two or more photons interfering is 1 in 9.27×10^{12} .

This result suggests that a single photon can interfere with itself as it passes through the slits. This agrees with the wavelike picture of the photon, as all waves possess this ability. However, each photon's detection occurs when it hits a localized area of the PMT with an energy related to that of the electron that is released. The detection of these waves, then, is in agreement with the particle-like picture of the photon, so they cannot be arriving as dispersed waves. Therefore, it must be the case that the photon passes through one slit or the other and reaches the detector as a particle. The only way to still produce an interference pattern then is if this particle has a spatial extent such that it can still interact with itself.

5 CONCLUSION

In this experiment, we measured the signal produced by photons when they pass through a single slit or a double slit configuration. Interestingly, whether there are multiple photons streaming through at once or only a single photon, the same double slit interference pattern is produced. Measuring this function with a laser provided precise physical measurements of the slit characteristics, where we found a slit width of $a = 0.071 \pm 0.001$ mm and separation of $b = 0.410 \pm 0.005$ mm. Using these measurements to model the single photon cases demonstrated the qualitative and quantitative agreement of these data.

This result demonstrates the fundamental paradox of quantum mechanics: the particle-wave duality of particles like photons. We showed that our results rely on photons displaying both of these behaviors, and that while the photon must behave like a particle it must also have a spatial extent that interacts with both slits.

This work can be improved by reducing errors in several parts of our analysis. First, when calibrating the instrument, we measured the photodiode signal as a function of detector position. However, there was a slight 0.01 V difference between the signal for the left and right halves of the box. This is of order the detector background, but clearly introduced a bias in both of our fits. Additionally, automating the micrometer would reduce errors twofold: (1) manually starting

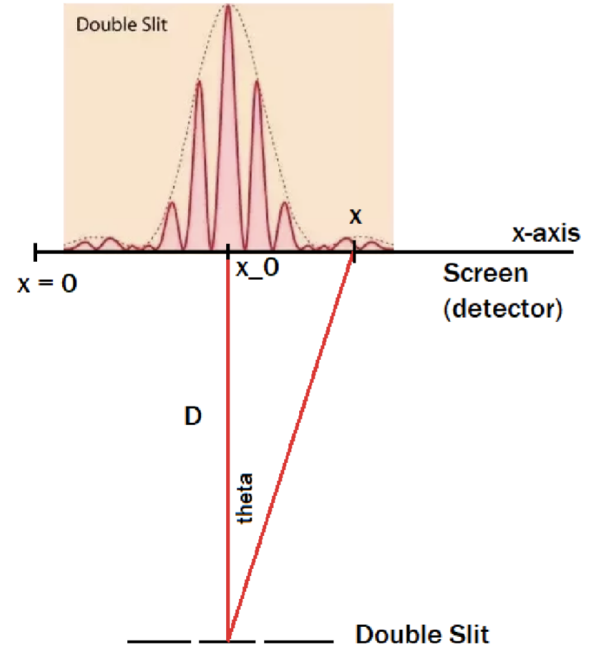


Figure A1. Top: diffraction pattern from a double slit apparatus. Bottom: Schematic of the geometry of the experiment.

and stopping is imprecise and subject to human errors and (2) an automatic procedure would mitigate position uncertainties to less than 0.005 mm.

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APPENDIX A: DERIVATION OF FRAUNHOFER INTENSITY EQUATION

The Fraunhofer equation models the intensity I of a wave as a function of angle θ from center. This solution assumes the wave arrives at the slit as a plane wave ([Meyer-Arendt 1984](#)). The intensity is a function of the phase displacement α :

$$I = I_0 \frac{(\sin \alpha)^2}{\alpha^2} \quad (\text{A1})$$

where $\alpha = \frac{\pi a \sin \theta}{\lambda}$. For the double slit, the solution is:

$$I = I_0 \frac{(\sin \alpha)^2}{\alpha^2} \cos^2(\beta) \quad (\text{A2})$$

where $\beta = \frac{\pi b \sin \theta}{\lambda}$.

In Figure A1, the geometry reveals that $\tan(\theta) = \frac{x - x_0}{D}$. Since

$\sin(\theta)$ is small compared to D , the small angle approximation is appropriate, so we have that $\sin(\theta) = \tan(\theta)$. Then α and β become

$$\alpha = \left(\frac{\pi a}{\lambda} \right) \left(\frac{x - x_0}{D} \right) \quad (\text{A3})$$

$$\beta = \left(\frac{\pi b}{\lambda} \right) \left(\frac{x - x_0}{D} \right) \quad (\text{A4})$$

We define new parameters A and B , where $A = \frac{\pi a}{\lambda D}$ and $B = \frac{\pi b}{\lambda D}$, such that:

$$I = I_0 \left(\frac{\sin(A(x - x_0))}{A(x - x_0)} \right)^2 \quad (\text{A5})$$

$$I = I_0 \left(\frac{\sin(A(x - x_0))}{A(x - x_0)} \right)^2 \cos^2(B(x - x_0)) \quad (\text{A6})$$

Thus, fitting these equations for the parameters A and B provides a simple calculation of a and b .

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